

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:35

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:

Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

1. Introduction

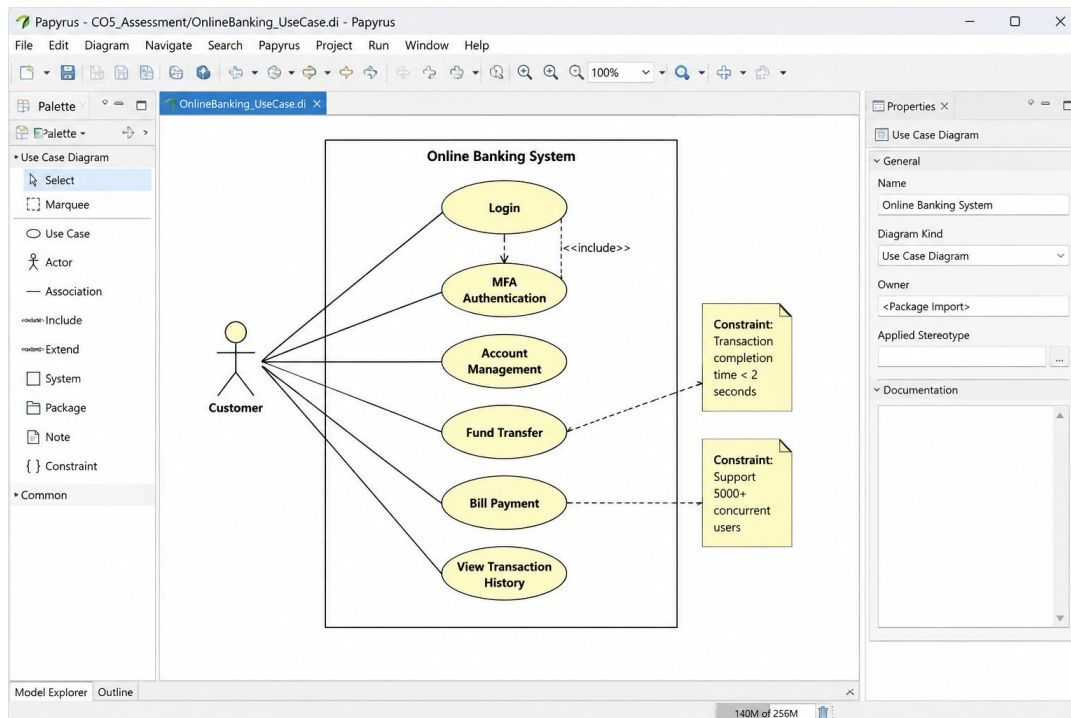
The Online Banking System consists of Account Management, Fund Transfer, Bill Payment, and Transaction History. The system must complete transactions within two seconds, provide multi-factor authentication, and support more than 5000 concurrent users.

2. Constraint Analysis

The transaction completion time constraint makes performance an important quality requirement. The requirement for multi-factor authentication makes security testing essential because unauthorized users must not be able to access customer accounts. The requirement to support more than 5000 concurrent users makes load and stress testing necessary to ensure that the system remains stable during heavy usage.

3. Potential Defects

From a security perspective, possible defects include unauthorized account access, weak authentication, bypassing multi-factor authentication, SQL injection, and unauthorized fund transfers. From a performance perspective, the system may experience slow transaction processing, delayed database responses, or reduced performance when a large number of users access the system simultaneously. From a reliability perspective, possible defects include duplicate transactions, incorrect account balances, transaction failures, and loss of transaction data during system failures.



4. Testing Strategies

Security testing should be performed to verify authentication, authorization, multi-factor authentication, session management, and protection against common security attacks. Performance testing should include load testing and stress testing with more than 5000 concurrent users to verify that transactions are completed within two seconds. Reliability testing should verify transaction rollback, database recovery, backup restoration, and correct account balances after failures. Concurrency testing should also be performed to ensure that multiple users can perform transactions without creating inconsistent data.

5. Justification

These testing techniques ensure that the banking system protects sensitive customer information, provides fast transaction processing, and continues to operate correctly under heavy workloads. Therefore, quality assurance helps make the system secure, performant, reliable, and suitable for real-world banking operations.

6. Conclusion

A combination of security, performance, reliability, and concurrency testing is required to satisfy the given constraints. These techniques ensure that the Online Banking System can provide safe and dependable services to a large number of users.

Q2. Scenario: E-Commerce Shopping Application

System Features:

Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation
- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

1. Introduction

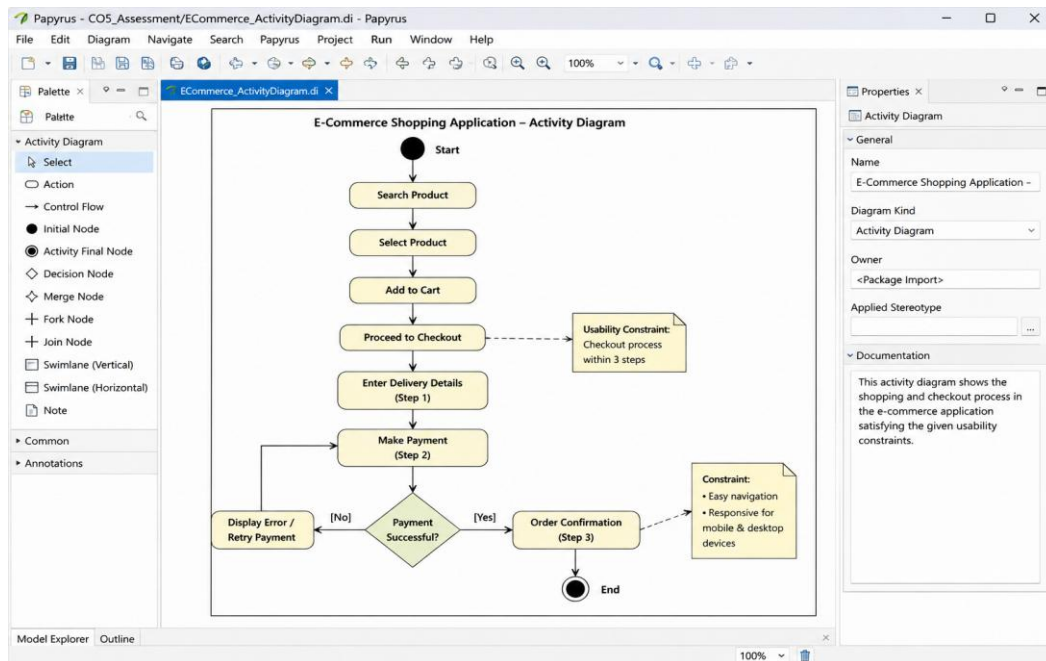
The E-Commerce Shopping Application consists of Product Search, Shopping Cart, Checkout, and Payment Gateway features. The application must provide a checkout process within three steps, allow easy product navigation, and work responsively on both mobile and desktop devices.

2. Constraint Analysis

The three-step checkout constraint requires the purchasing process to be short and simple. Easy product navigation requires users to find products without unnecessary effort. The responsive design constraint requires the interface to adapt properly to different screen sizes and devices.

3. Usability Challenges

Users may experience difficulties if the product navigation is complicated or if the search option is difficult to locate. A checkout process containing too many pages can increase user effort and cause users to abandon their purchases. Small buttons, poorly arranged forms, and unclear payment instructions can create additional problems for mobile users. Poorly written error messages can also make it difficult for users to correct their mistakes.



4. Usability Testing

User testing should be performed by asking users to search for products, add them to the shopping cart, and complete a purchase. Task completion testing can be used to measure whether users can complete the checkout process within three steps. Navigation testing can determine whether users can easily move between product search, shopping cart, and checkout. Responsive testing should be performed on different mobile and desktop screen sizes to ensure that all interface elements remain usable.

5. Recommended Improvements

The application should provide a clearly visible search bar and simple product categories to make navigation easier. The checkout process should be reduced to three simple steps so that users can complete their purchases quickly. Buttons, labels, and forms should be clearly displayed, especially on mobile devices. Meaningful error messages should be provided whenever users enter incorrect information.

6. Justification

These improvements reduce the amount of effort required from users and decrease the number of errors during shopping. A simple checkout process and easy navigation improve efficiency, while responsive design provides a consistent experience across different devices.

7. Conclusion

Usability testing helps identify problems in the shopping application before deployment. By simplifying navigation, limiting checkout to three steps, and providing responsive design, the application can provide better customer satisfaction and efficiency.

Q3. Scenario: Hospital Management System

Modules:

Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability.

(BL5)

1. Introduction

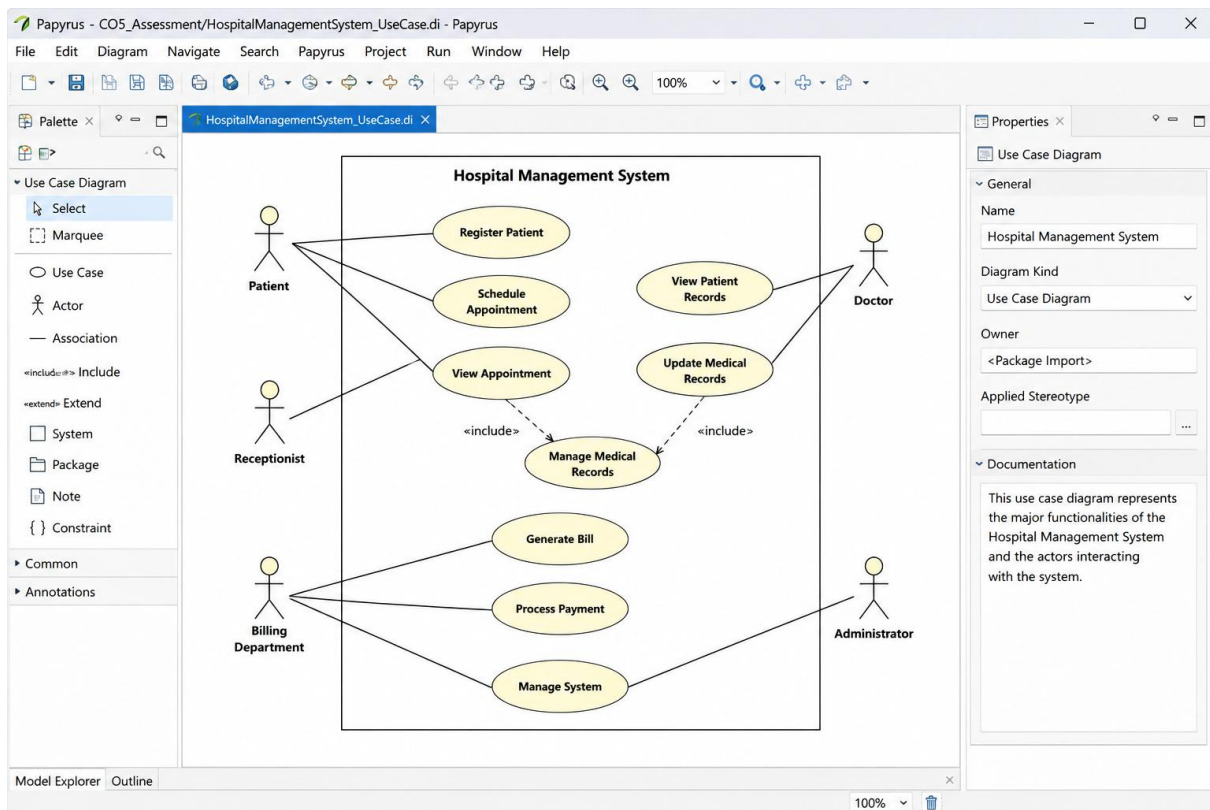
The Hospital Management System consists of Patient Registration, Appointment Scheduling, Medical Records, and Billing modules. The system must maintain accurate patient data, remain available 24 hours a day and seven days a week, and support simultaneous access by multiple departments.

2. Constraint Analysis

Patient data accuracy is the most important requirement because incorrect information can affect hospital operations. The 24/7 availability requirement means that the system should continue operating with minimum downtime. The simultaneous access requirement means that multiple departments must be able to access and update information without causing data conflicts.

3. Critical Test Cases

Patient registration should be tested using valid and invalid patient information, and the system should prevent duplicate registrations. Appointment scheduling should be tested for booking, cancellation, rescheduling, and prevention of double bookings. Medical records should be tested to ensure that the correct patient information is created, retrieved, and updated. Billing should be tested to verify correct bill calculations, payment processing, and prevention of duplicate billing.



4. Quality Assurance Practices

Functional testing should be performed to verify that every hospital module works according to its requirements. Data validation testing should ensure that patient information is accurate and complete. Security testing should ensure that only authorized personnel can access medical records. Concurrency testing should verify that multiple departments can access the system simultaneously without causing inconsistent data. Availability testing should verify that the system remains operational continuously, while recovery testing should verify that the system can recover correctly after hardware, software, or database failures.

5. Testing Priorities

Testing should give the highest priority to patient data accuracy because incorrect data can have serious consequences. System availability should receive the next priority because hospital services depend on continuous access to information. Concurrent access should also be tested carefully because several departments may work with the same patient information at the same time.

6. Justification

Proper quality assurance prevents incorrect patient information, appointment conflicts, billing errors, and unauthorized access. Concurrency and recovery testing also ensure that the system remains reliable when many departments use it simultaneously or when failures occur.

7. Conclusion

A strong quality assurance process ensures that the Hospital Management System remains accurate, secure, continuously available, and reliable. Testing the system according to its constraints helps prevent critical defects and improves overall system reliability.

Q4. Scenario: Smart Library Management System

Features:

Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

1. Introduction

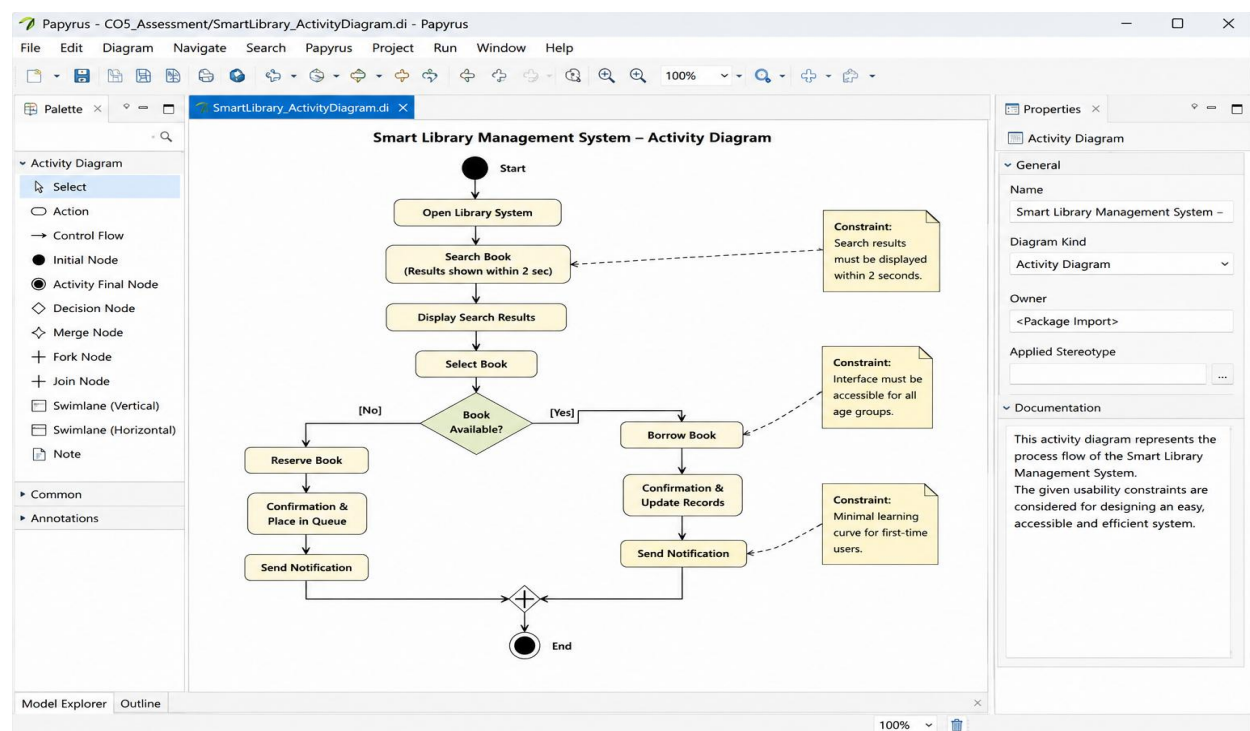
The Smart Library Management System provides Book Search, Book Reservation, Borrow/Return, and Notification features. The system must display search results within two seconds, provide an accessible interface for all age groups, and have a minimal learning curve for first-time users.

2. Constraint Analysis

The two-second search requirement makes system performance an important usability factor. Accessibility for all age groups requires readable text, clear controls, and simple navigation. The minimal learning curve requires the interface to be simple enough for first-time users to understand without extensive instructions.

3. Usability Challenges

Users may experience difficulty if the search process is slow or if the navigation structure is complicated. Small fonts, unclear icons, poor colour contrast, and complex menus may create accessibility problems for some users. First-time users may also have difficulty understanding the system if instructions and labels are unclear.



4. Usability Testing Methods

User testing should be performed by asking users to search for a book, reserve it, borrow it, and return it. Task completion testing should measure how quickly and accurately users can perform

these activities. Accessibility testing should verify font readability, colour contrast, button size, labels, and ease of navigation. Performance testing should verify that search results are displayed within two seconds.

5. Recommended Improvements

The system should provide a simple and clearly visible search box so that users can quickly find books. The interface should use readable fonts, clear labels, familiar icons, and sufficient colour contrast. Navigation should be kept simple, with unnecessary menus and options removed. Clear instructions and meaningful error messages should be provided to help first-time users understand the system.

6. Justification

These improvements make the system easier to understand and operate. Faster search results improve efficiency, while accessible design allows users from different age groups to use the system comfortably. Simple navigation also reduces the learning time for new users.

7. Conclusion

Usability testing ensures that the Smart Library Management System is fast, accessible, and easy to learn. The proposed improvements directly satisfy the given usability constraints and improve the overall user experience.

Q5. Scenario: Airline Reservation System

Modules:

User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic
- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**

1. Introduction

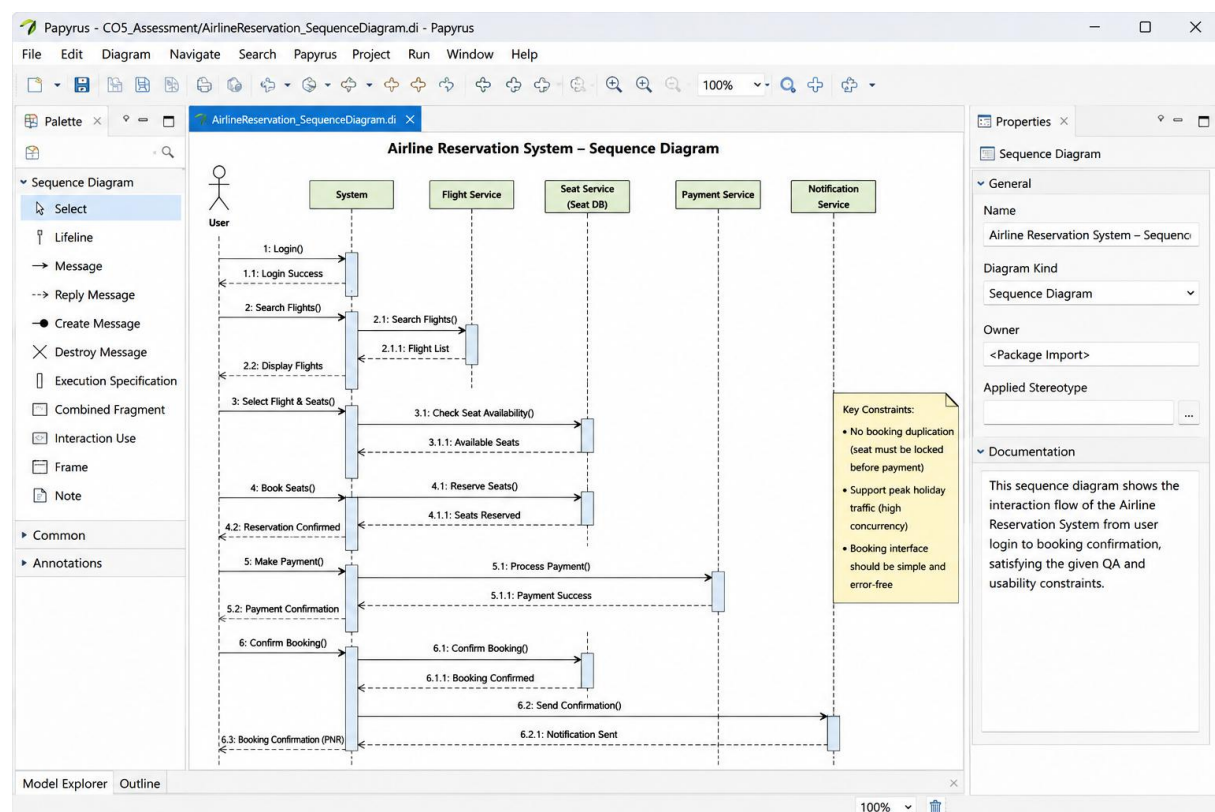
The Airline Reservation System consists of User Login, Flight Search, Seat Selection, and Ticket Booking modules. The system must prevent duplicate bookings, support peak holiday traffic, and provide a simple and error-free booking interface.

2. Constraint Analysis

Preventing duplicate bookings is important because two users must not be allowed to reserve the same seat. Supporting peak holiday traffic requires the system to handle a large number of simultaneous users without performance degradation. A simple and error-free interface is necessary to reduce user mistakes during flight and seat booking.

3. Major Risks

The major reliability risk is duplicate seat booking, which can occur when multiple users attempt to reserve the same seat simultaneously. Performance risks include slow response times and server overload during peak holiday periods. Security risks include unauthorized account access and exposure of passenger information. Usability risks include complicated forms, confusing seat-selection interfaces, and unclear error messages.



4. Quality Assurance Testing

Functional testing should verify flight searching, seat selection, ticket booking, and cancellation operations. Concurrency testing should be performed by allowing multiple users to attempt to book the same seat simultaneously. Load and stress testing should verify that the system can handle peak holiday traffic. Security testing should verify login protection and unauthorized access prevention.

5. Usability Testing

Usability testing should evaluate whether users can easily search for flights and select seats. The passenger information form should be tested to ensure that users can enter information without confusion. The system should provide clear error messages when incorrect information is entered.

6. Optimization

The system should use proper database transaction management and seat locking mechanisms to prevent duplicate seat bookings. Load balancing and performance optimization should be used to handle large numbers of users during holiday periods. The booking interface should use simple forms, clear labels, and understandable error messages.

7. Justification

Quality assurance testing improves the reliability, security, and performance of the reservation system, while usability testing ensures that users can complete bookings easily and accurately.

8. Conclusion

Applying both quality assurance and usability testing ensures that the Airline Reservation System remains reliable during peak traffic and prevents booking errors. A simple interface combined with proper concurrency and performance testing provides a secure, efficient, and user-friendly reservation system.